

Recommender System (CS4053)

Sessional-I Exam

Date: 2/24/2025

Course Instructor

Syed Zain Ul Hassan

Total Time: 1hr

Total Marks: 30

Total Questions: 03

Roll No

Section

Student Signature

Do not write below this line.

Attempt all questions.

CLO # 1: Describe different techniques in making personalized recommendations in various scenarios.

Q1 (a): Consider the following interaction table:

[Estimated Time: 10 minutes, Points: 6]

	Product 1	Product 2	Product 3	Product 4	Product 5
User 1	1	2	?	3	5
User 2	2	?	1	4	?
User 3	?	1	2	2	5
User 4	5	3	3	1	5

Find the missing rating $R(\text{User2}, \text{Product 2})$ using Item-based Collaborative Filtering with mean-centered prediction function and Cosine Similarity as similarity metric. Assume $k=2$.

Q1 (b): Based on observation, which one of the given products in Part (a) appears to be affected by a shilling attack? Justify your answer.

[Estimated Time: 5 minutes, Points: 4]

CLO # 1: Describe different techniques in making personalized recommendations in various scenarios.

Q2 (a): Consider the following interaction table:

[Estimated Time: 15 minutes, Points: 6]

	Product 1	Product 2	Product 3	Product 4
User 1	1	2	2	3
User 2	?	?	1	1
User 3	2	?	2	?
User 4	3	2	3	1

Use User-based Naïve Bayes Collaborative Filtering to predict the missing rating $R(\text{User 3}, \text{Product 2})$. Assume $\alpha = 0.01$.

Q2 (b): Consider a website with 10k subscribers and 230k movies. If the rating system offered to users has a range of 1-10. Would it be feasible to use a probabilistic collaborative filtering approach in this case? Justify your answer.

[Estimated Time: 5 minutes, Points: 4]

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CLO # 2 Solve mathematical optimization problems pertaining to recommender systems.

Q3 (a): Consider the given matrix:

[Estimated Time: 15 minutes, 6 points]

	Features	Rating
Item 1	f_1, f_3, f_4	4
Item 2	f_2, f_1, f_3	?
Item 3	f_1, f_2, f_4, f_5	?
Item 4	f_5, f_3, f_4	?
Item 5	f_1, f_5, f_2, f_3	3

The active user has already rated Item 1 and Item 5. Based on the given features, which one of the three unrated items will be recommended to the user? The allowed values for ratings are 1, 2, 3, 4 and 5.

Q3 (b): Consider a set of products with 10k features and the total number of products in the system are 1k with 1000k users. Is Content-based Filtering a suitable approach to be used in this?

[Estimated Time: 5 minutes, Points: 4]

Good luck